

CS & IT ENGINEERING

Algorithm

Miscellaneous

Lecture No. 2



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Topic

TC analysis using
Precedence Tree.

Topics to be Covered



Topic

Topic

Topic

OBST

Optimal Cost BST



About Aditya Jain sir



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.





Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

Sunday → 3rd Aug.

1) 8:15 AM - 10:15 AM

2) 11:30 AM - 1:30 PM



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Binary Search Tree : (BST)

BST is a Binary Tree, which has the property that the value at every node is less than the value of the right child (and all the values in the right sub-tree) and the value at every node is greater than the value of the left child (and all the values in the left sub-tree).



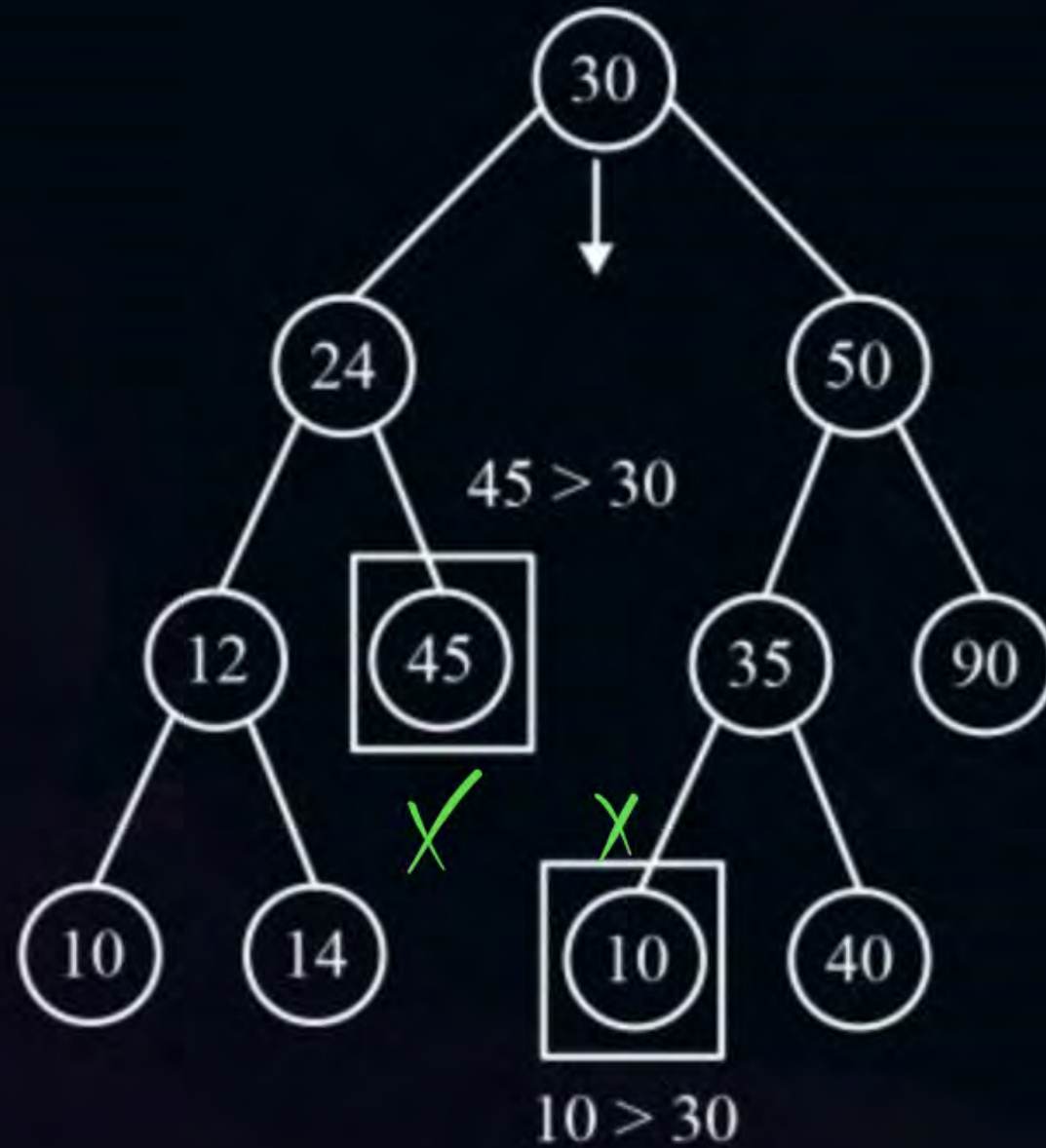
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Equation:

BT ✓

BST ✗





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Optimal cost binary Search tree (OBST)

#Q. For given data, which of the following Binary Search Trees (BSTs) will result in minimum cost? (Optimization Problem)

A green arrow originates from the underlined text '(Optimization Problem)' and points towards the left.

[NAT]



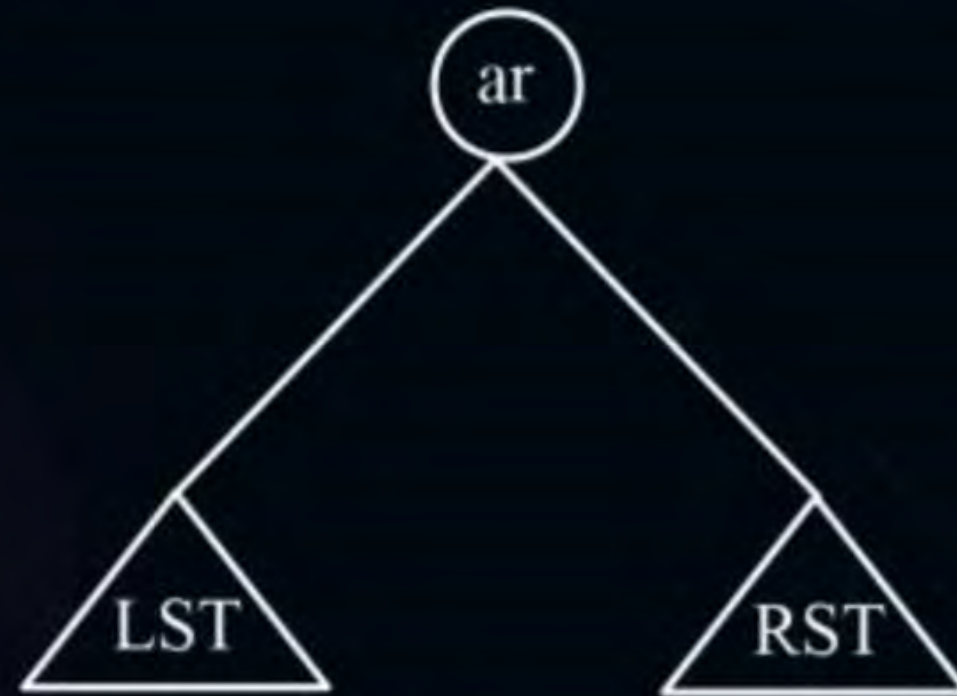
#Q. Construct the Binary Search Tree with minimum cost → DP based recurrence



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BST → General representation



All values in LST

Q_r
~~< ar <~~

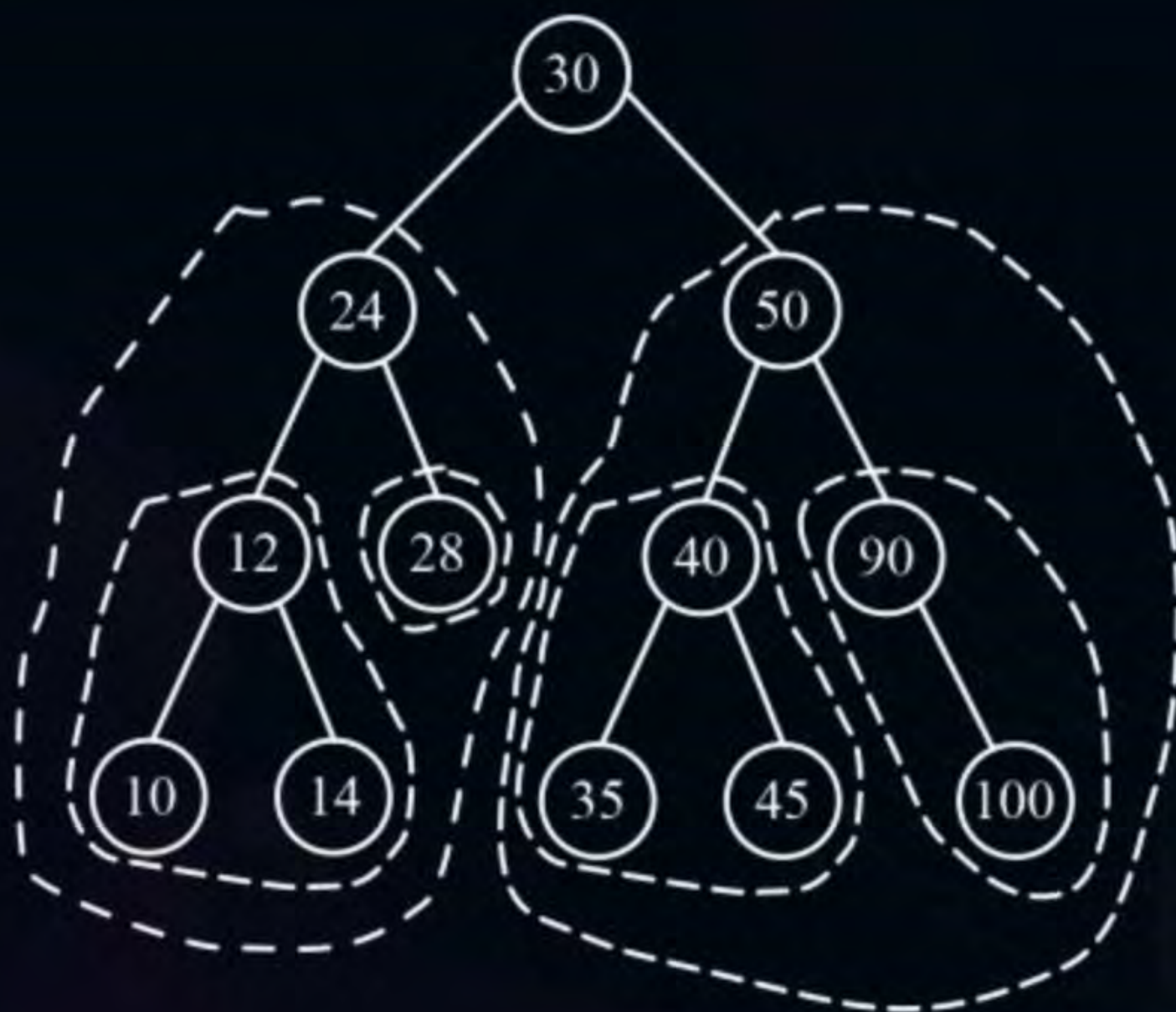
All values in RST



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Equation:



Valid BST



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A Binary Search Tree for words / Text / Alphabets
(lots of applications in Computer Science)

Idea: Here comparison is made on the basis of lexicographical order. → (dictionary order)



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~~Equation.~~

eg :-

$$a < b$$

$$ab < ad$$

$$a = a$$

$$b < d ?$$

↓ ↓
ab < ad
↑ ↑

$$Adit > Aditya$$

→ False

$$a = a$$

$$ad = ad$$

$$adi = adi$$

$$adit = adit$$

$$adit < aditya$$



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Equation-3:

zebra < Aditya jain → False

z > a

abd > acz → False

b > c (✗)

z > a



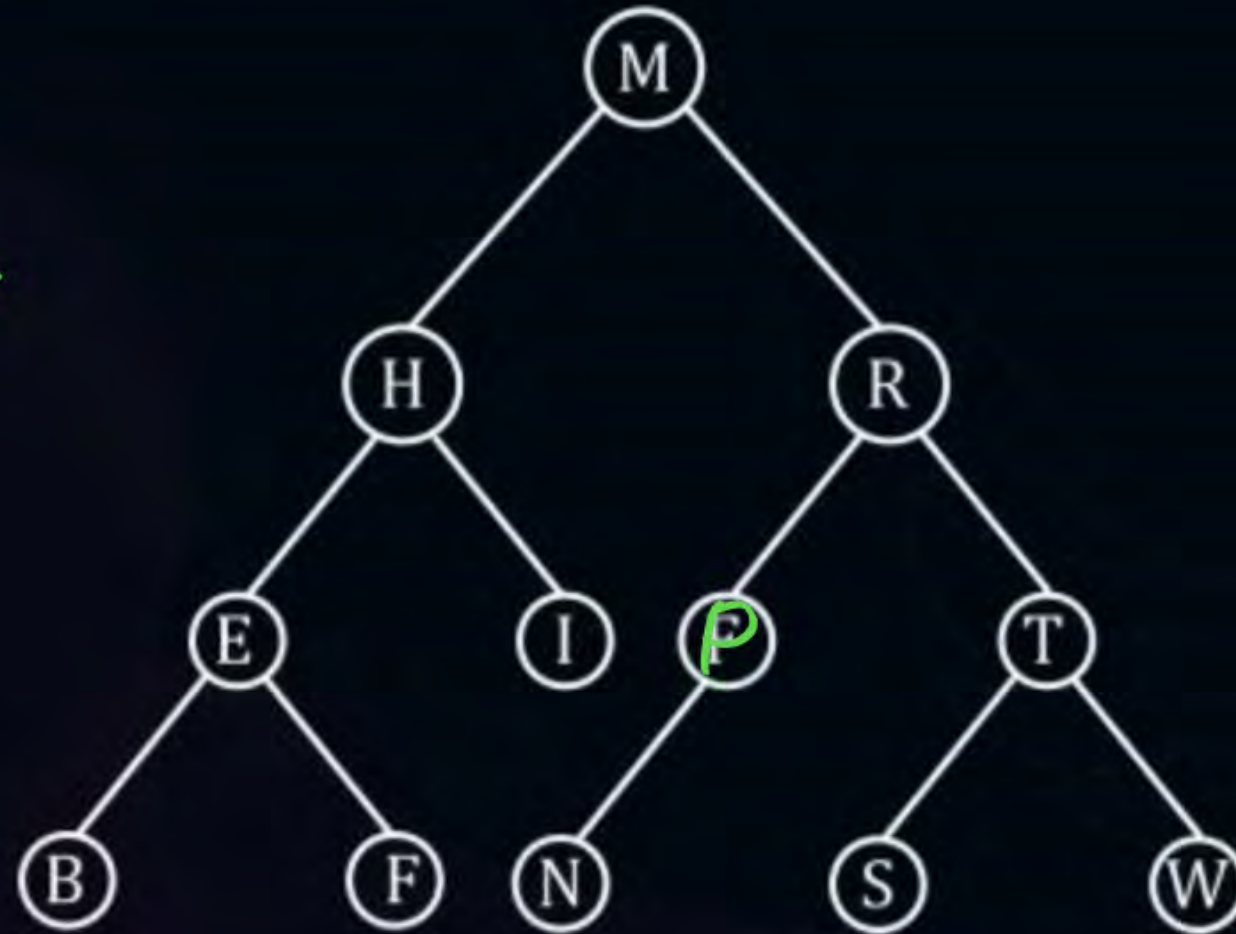
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Equation-2:

Is this a valid BST?

Yes





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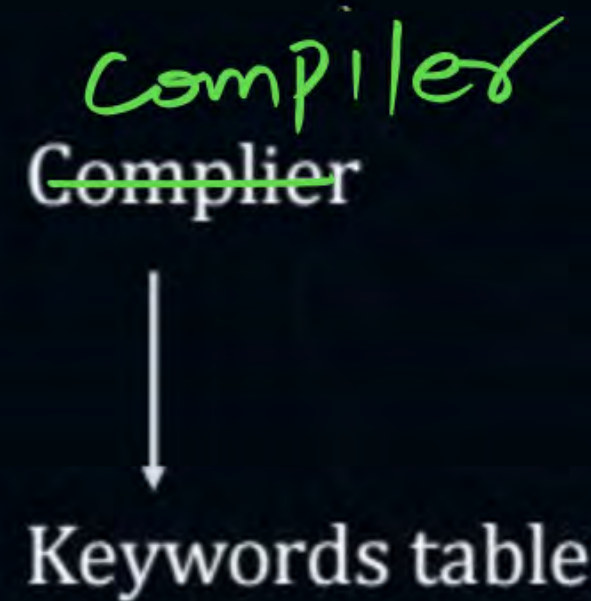


Application of BSTs (in computer Science)

1. Efficient Searching
2. File/ data organization
3. Indexing

Keywords:

For, while, if, else





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In context~~x~~ of complies /programming language:

Keywords

if, else

for, while

break

continue

int

return

Non-keywords

i, a,b aj,

first_name

age

[NAT]



#Q. For n distinct elements, how many different binary search tree are possible?



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Catalan number: (n)

$$\text{Number of BSTs} = \frac{1}{(n+1)} * {}^{2n}C_n$$



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Equation:

$$n = 3$$

$$n \frac{1}{(n+1)} *^{2n} C_n$$

$$\frac{1}{(3+1)} *^6 C_3$$

$$= \frac{1}{4} \times \frac{6!}{3! \times 3!}$$

$$= \frac{1}{4} \times \frac{6 \times 5 \times 4 \times 3!}{3! \times 3!} = 5$$



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Example:

$A = [a_1, a_2, a_3] = [\text{do}, \text{if}, \text{while}]$

$\text{do} < \text{if} < \text{while}$

Number of difference BSTs?

Number of identifiers $n = 3$

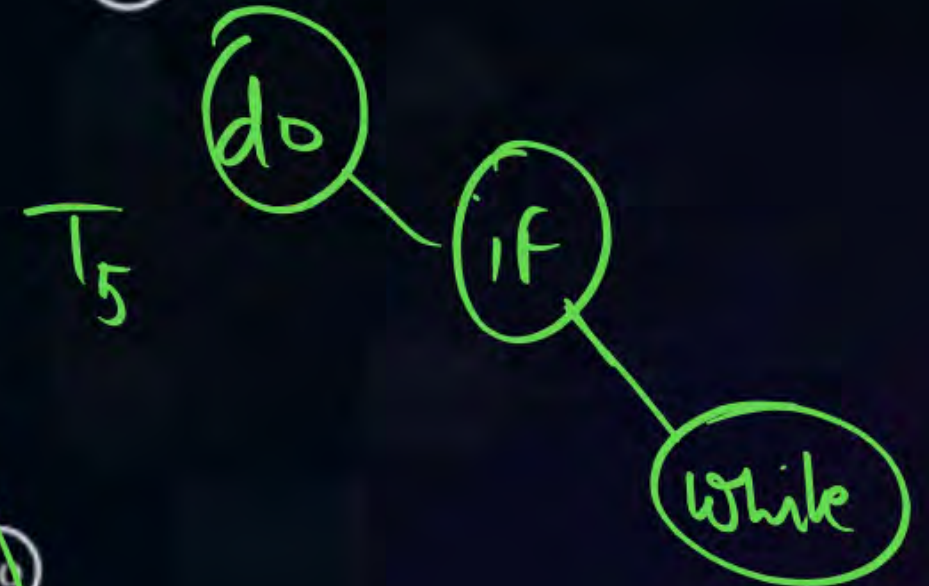
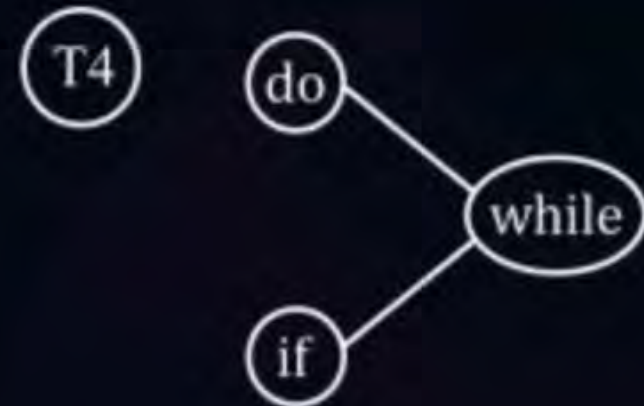
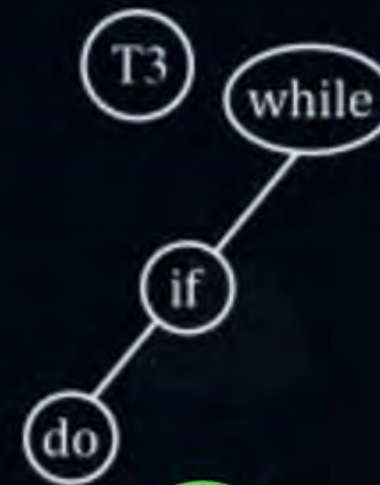
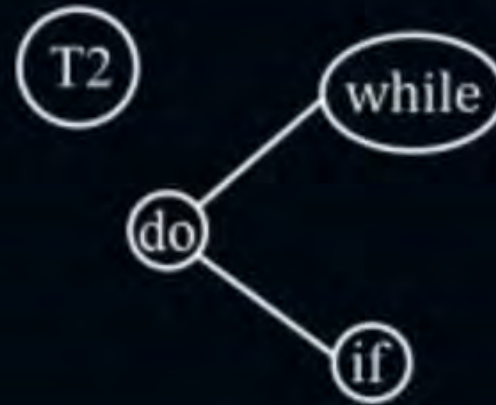
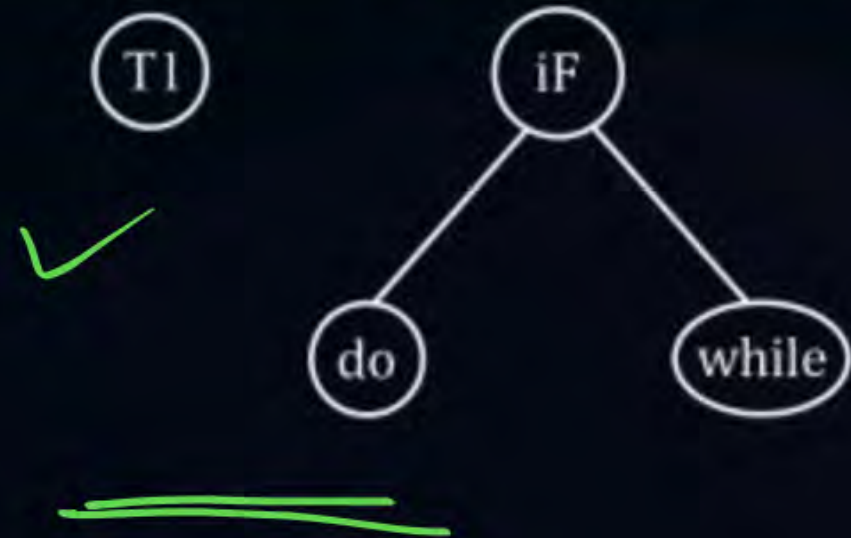


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~~Basic infection~~

Prefer T1 as it has lower depth (min cost)





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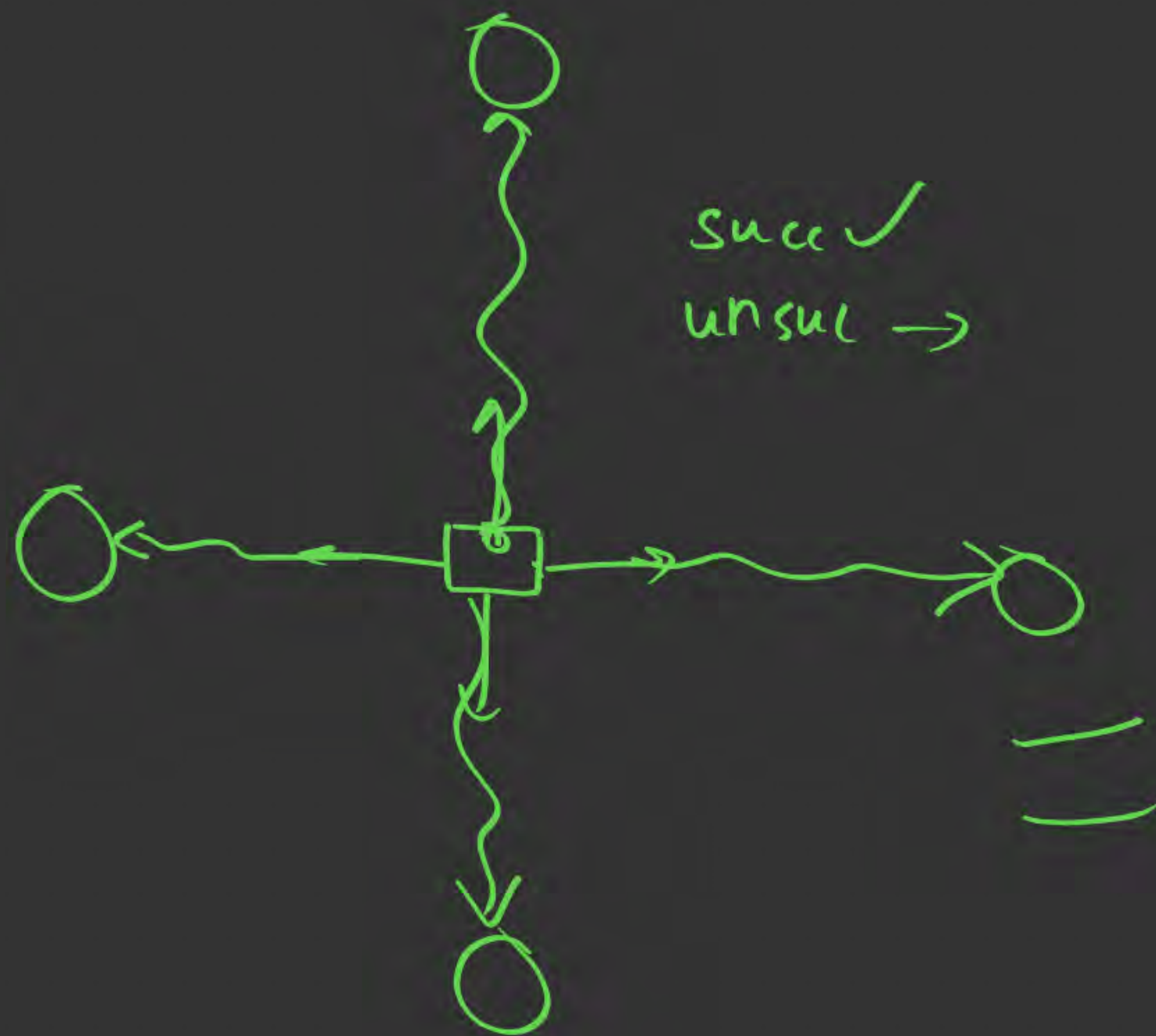
Process to determine the cost of BST (T):

1. $\text{Cost (T)} = \text{cost (successful search)} + \text{cost (un-successful search)}$



- A successful search will always end at a given tree node (non-null link)
- An un-successful search will always end at a null link.

Mr. Nobody





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Note: Throughout the discussion, the assumption is that root node is at level. 1 .



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a_i

$N \rightarrow$ identifiers

1. Cost (successful search)

$$= \text{cost}(a_1) + \text{cost}(a_2) \dots \text{Cost}(a_n)$$

$$= \sum_{i=1}^n \text{cost}(a_i)$$

2. Cost (a_i) $\propto$ level (a_i)

$$\text{Cost}(a_i) = P_i * \text{level}(a_i)$$

Where,

P_i = the probability or the usefulness of this identifier a_i in the application.



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Example:

~~if, confines~~ *if, continue*

Usually, in programming 'if' is used more frequently than 'continue'.

[NAT]



#Q. In a binary tree with n nodes, how many null links are there?



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Solution:

every node $\Rightarrow$ ~~2 links~~ *links*

Total ~~Total~~ links = $2 * n$

Tree with ^{*n*} nodes, has $(n-1)$ non-null links

Hence, null links = Total - (non-null links)

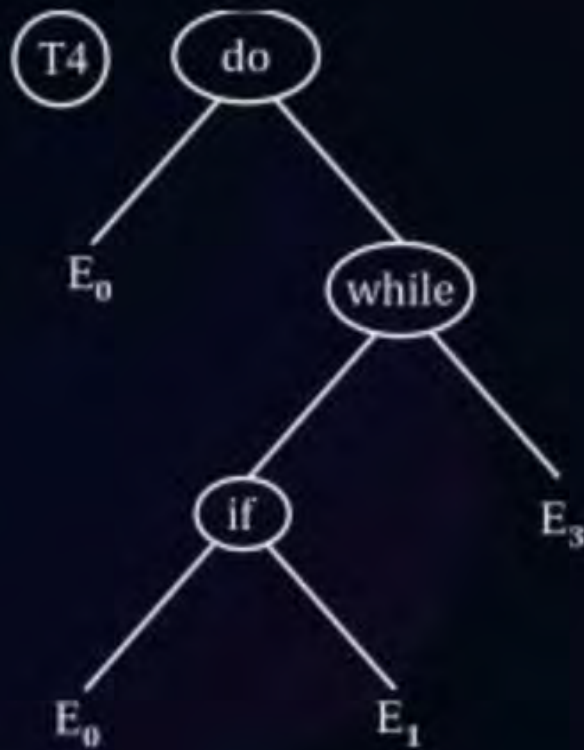
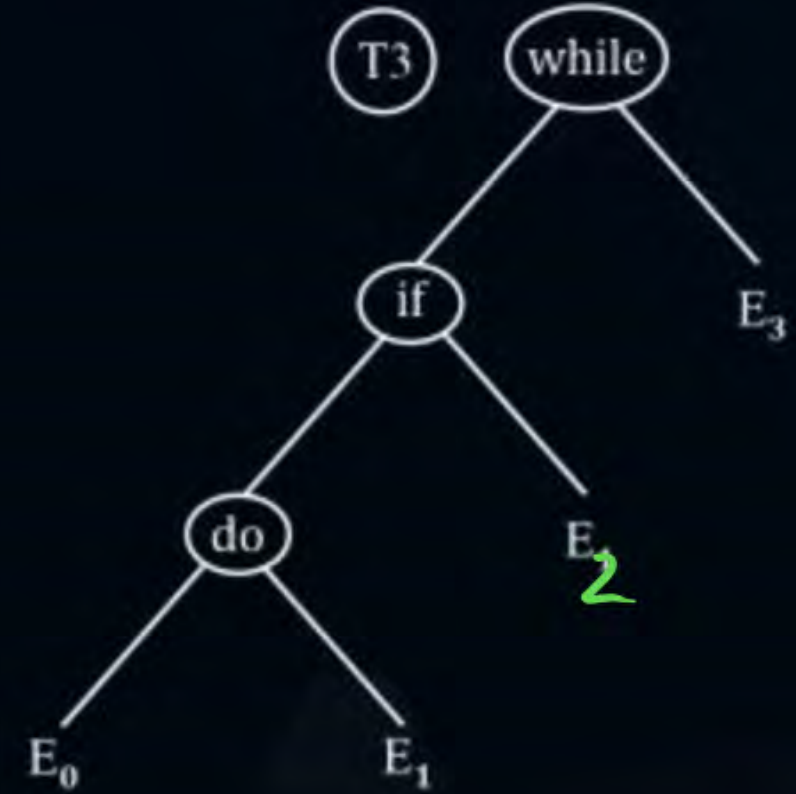
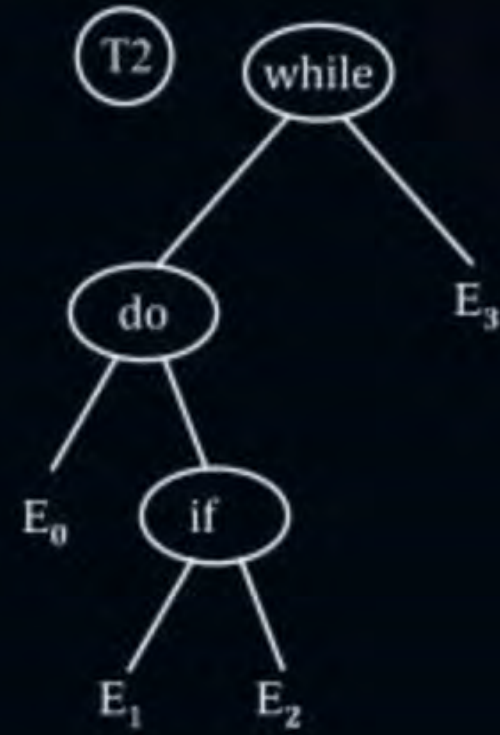
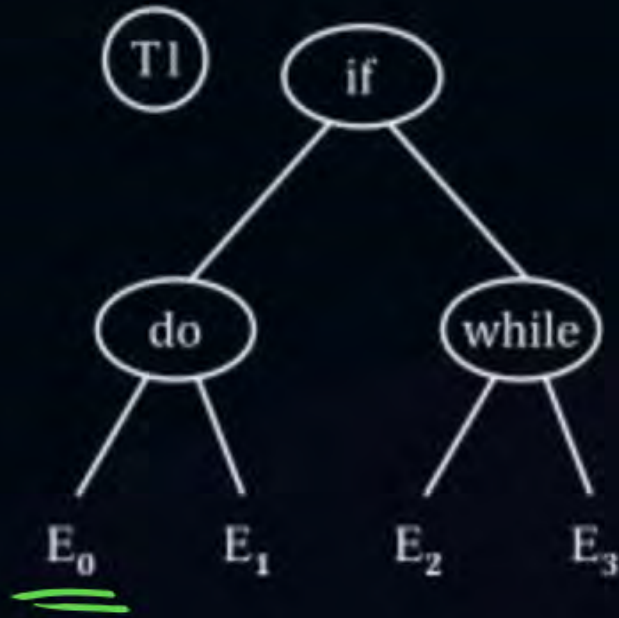
$$= 2n - (n-1)$$

$$= n + 1$$





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N nodes $\rightarrow$ $(n+1)$ null links $(E_0 \rightarrow E_n)$

1. Cost(unsuccesful search) = cost (E_0) + cost (E_1) + cost (E_n)

$$= \sum_{i=0}^n \text{cost}(E_i)$$

2. Cost $(E_i) \propto$ (Level $(E_i) - 1)$

$$\text{cost}(E_i) = q_i * (\text{level}(E_i) - 1)$$

$q_i \rightarrow$ probability of E_i (non-keyword)



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Example:

Write a programme to add y integers.

1. Programme / Books / Tutorials → general pattern.

int a, b, c, d

2. Define a for loop

For (i = 1, i <= n ; i ++)

For (aj = 1 ; aj < n; aj ++)

Pr (i) > Pr (aj)





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Summary: Cost of BST (T)

Cost (T) = Cost (successful search) + cost (un-successful search)

$$\text{Cost (T)} = \sum_{i=1}^n p_i * \text{level}(a_i) + \sum_{i=0}^n q_i * (\text{level}(E_i) - 1)$$

for solving questions:

Given:

$$q_1 \rightarrow n$$

$$P_1 \rightarrow n$$

$$q_0 \rightarrow n$$



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Given:

$$A = [a_1, a_2, a_3] = [\text{do}, \text{if}, \text{while}]$$

$$[p_1, p_2, p_3] = [0.5, 0.1, 0.05]$$

$$[q_0, q_1, q_2, q_3] = [0.15, 0.1, 0.05, 0.05]$$

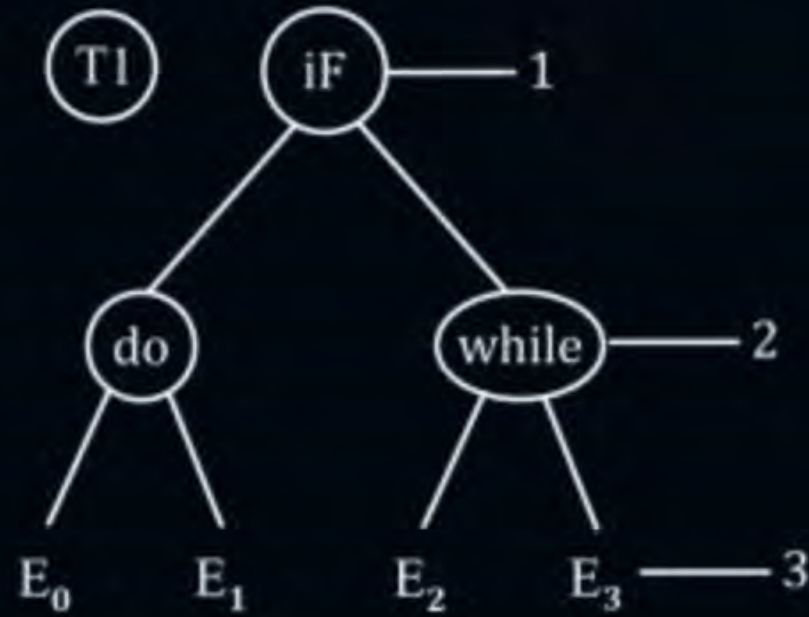
$$\text{Cost}(T_1) = \sum_{i=1}^n p_i * \text{level}(a_i)$$

+

$$= \sum_{i=0}^n q_i * (\text{level}(E_i) - 1)$$



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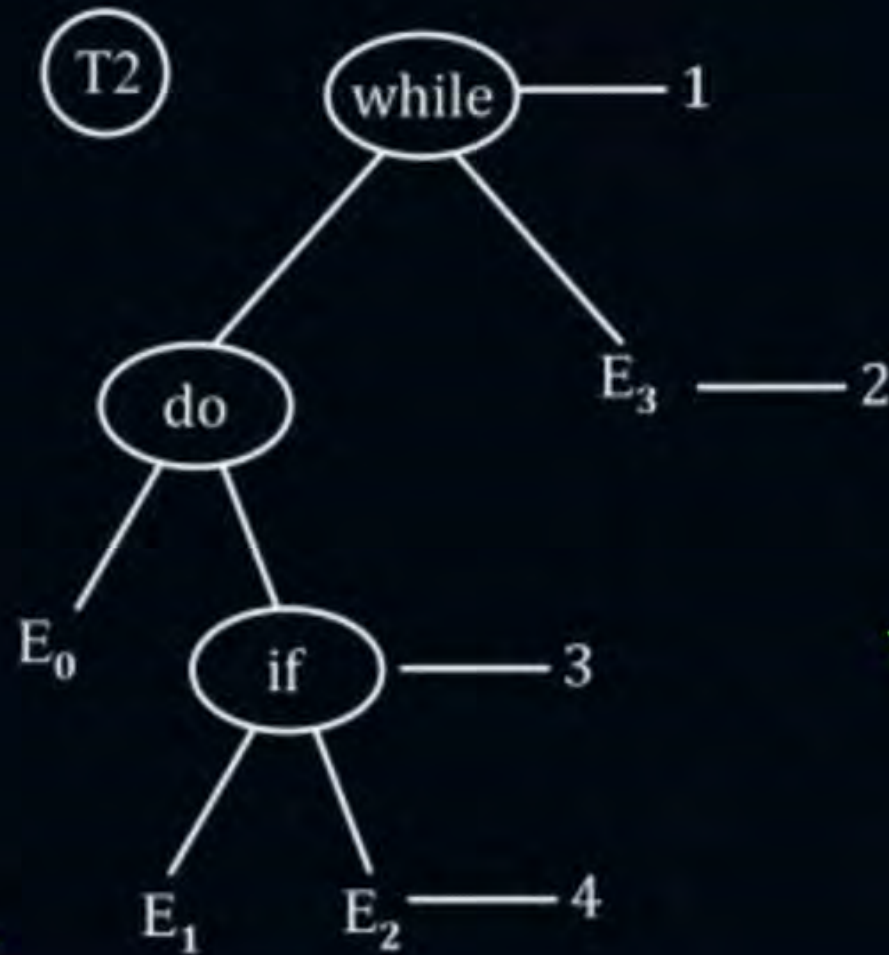


$$\text{Cost} = \underline{\underline{1.9}}$$

$$\begin{aligned} &= (0.5 \times 2 + 0.1 \times 1 + 0.5 \times 2) + (0.15 \times (3-1) + 0.1 \times (3-1) + 0.05 (3-1) + 0.05 (3-1)) \\ &= 1 + 0.1 + 0.1 + 0.3 + 0.2 + 0.1 + 0.1 \\ &= \underline{\underline{1.2}} + \underline{\underline{0.7}} = \underline{\underline{1.9}} \end{aligned}$$



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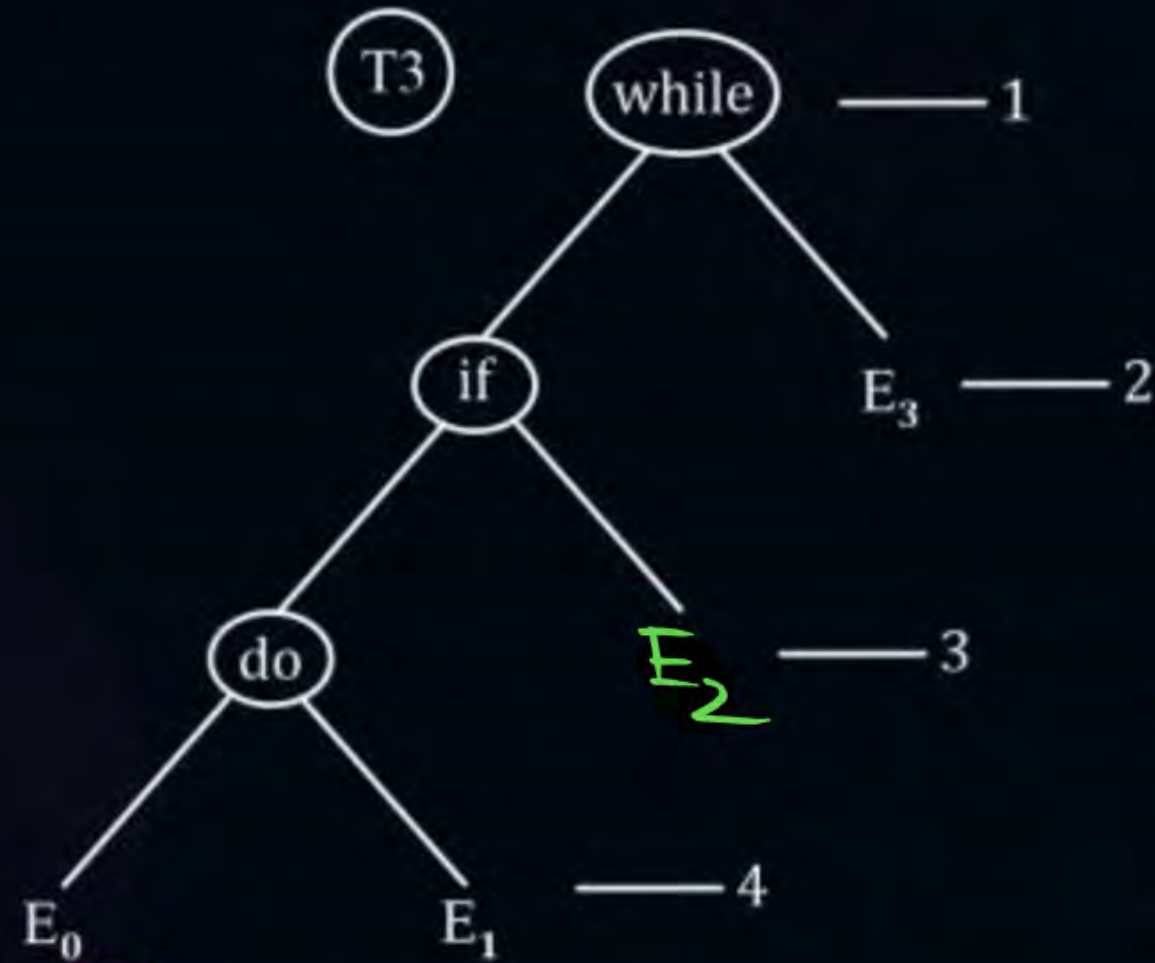


$\Rightarrow$ 2.15

$$\begin{aligned} &= [0.5 \times 2 + 0.1 \times 3 + 0.05 \times 1] + [0.15 \times (3-1) + 0.1 \times (4-1) + 0.05(4-1) + 0.05(2-1)] \\ &= (1 + 0.3 + 0.05) + (0.3 + 0.3 + 0.15 + 0.05) \\ &= \underline{2.15} \end{aligned}$$



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2.65

$$\begin{aligned} & [0.5 \times 3 + 0.1 \times 2 + 0.05 \times 1] + [0.15 \times (4-1) + 0.1 \times (4-1) + 0.05 \times (3-1) + 0.05 \times (2-1)] \\ &= (1.5 + 0.2 + 0.05) + (0.45 + 0.3 + 0.1 + 0.05) \\ &= 2.65 \end{aligned}$$



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Cost (T4)= ?

$$= (0.5 \times 1 + 0.1 \times 3 + 0.05 \times 2) + 0.15 \times (2-1) + 0.1 \times (4-1) + 0.05 (4-1) + 0.05 (3-1)$$

$$= 1.6$$



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Cost (T5) = ?

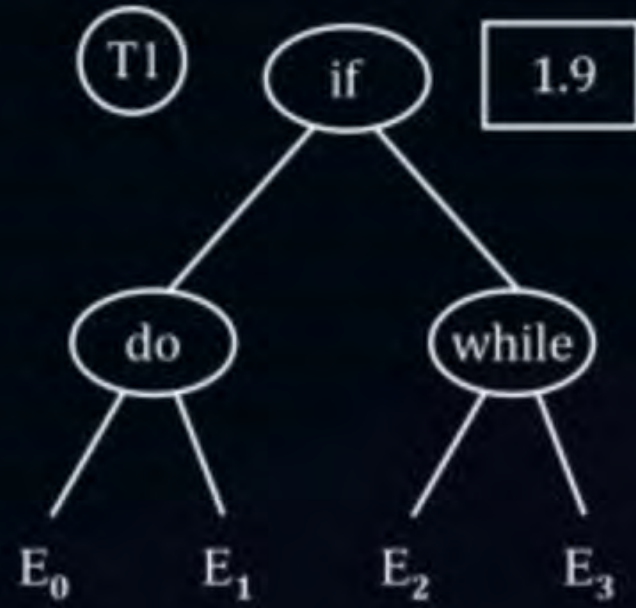
$$= (0.5 \times 1 + 0.1 \times 2 + 0.05 \times 3) + (0.15 \times (2-1) + 0.1 \times (3-1) + 0.05 \times (4-1) + 0.05 \times (4-1))$$

$$= (0.5 + 0.2 + 0.15) + (0.15 + 0.2 + 0.15 + 0.15)$$

$$= 1.5$$

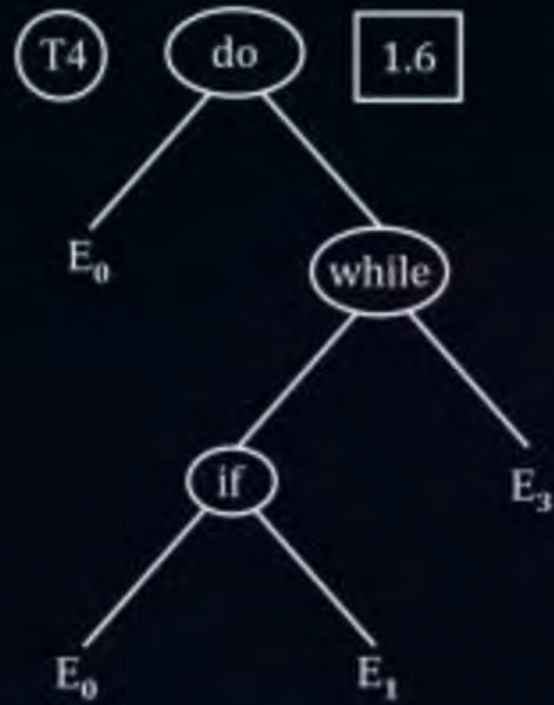


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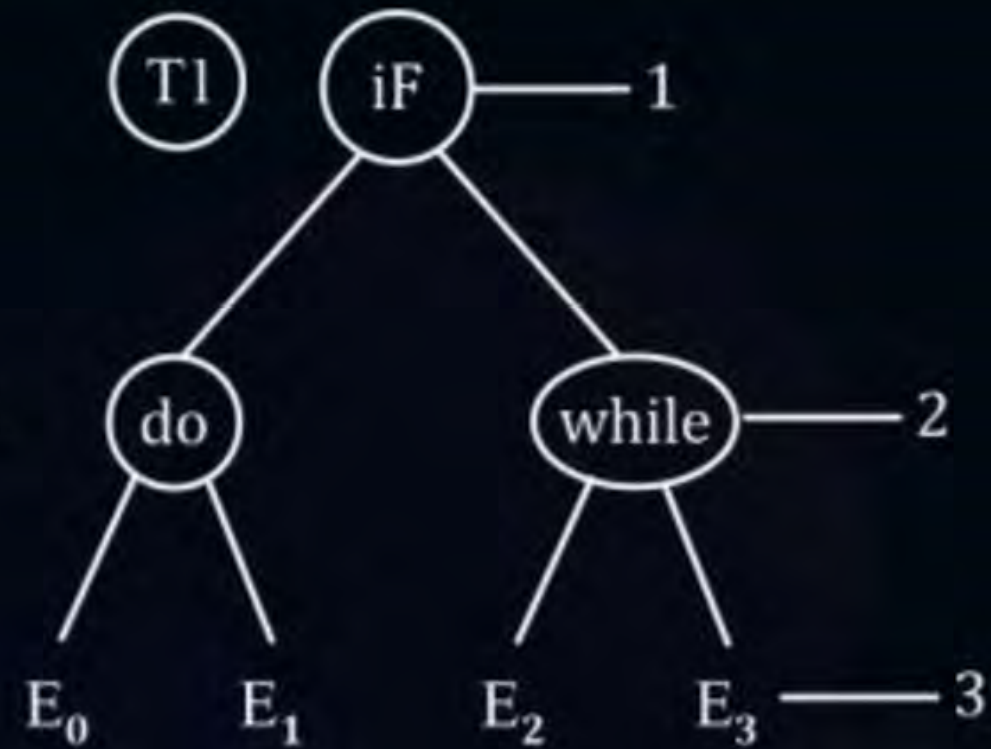




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Observation:



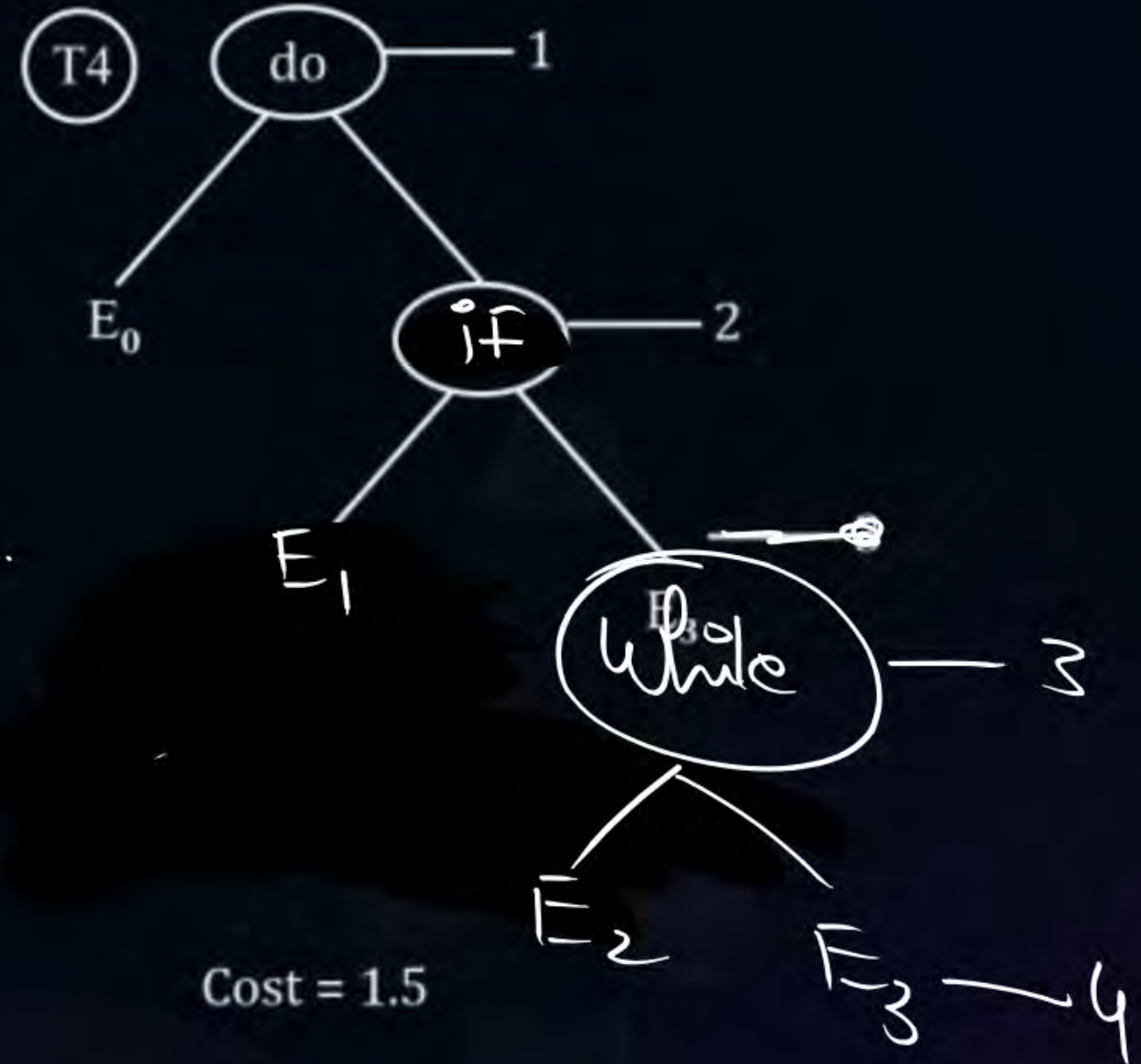
Cost = 1.9

do → 0.5

If → 0.1

While → ~~0.1~~ 0.05

$P_i \times \text{level}(a_i)$



Cost = 1.5



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Imp:-

The cost of the BST depends on both:

The depth (levels) as well as the probabilities of all the identifiers ~~$(p_i \& q_i)$~~ $(p_i \& q_i)$

Idea is very similar to Huffman Encoding / Optimal Merge Pattern.



2 mins Summary



→ OBST



THANK - YOU